

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 11: Tuesday 15 September 2026

(Week 6)

Remark. Q10.2.5, Q10.4 to Q10.9 are leftover exercises from the Friday class in Week 5.

Remark. Q11.1 to Q11.4 are on Section 13.8.

More questions on Section 13.8 will be set for Friday.

Remark. For a function $\mathbf{g}(\mathbf{x}, \mathbf{y})$, its graph is the surface $\mathbf{S}$ defined by the equation $\mathbf{z} = \mathbf{g}(\mathbf{x}, \mathbf{y})$. Note that

$$\mathbf{z} = \mathbf{g}(\mathbf{x}, \mathbf{y}) \iff \mathbf{g}(\mathbf{x}, \mathbf{y}) - \mathbf{z} = 0.$$

Thus $\mathbf{S}$ can also be regarded as the level surface

$$\mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 0, \quad \text{where } \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \mathbf{g}(\mathbf{x}, \mathbf{y}) - \mathbf{z}.$$

Q10.2.5. (Supplementary Exercise) Recall that the equation of the tangent plane to the graph $\mathbf{z} = \mathbf{g}(\mathbf{x}, \mathbf{y})$ at $(\mathbf{x}_o, \mathbf{y}_o)$ is given by

$$\mathbf{z} = \mathbf{g}(\mathbf{x}_o, \mathbf{y}_o) + \mathbf{g}_x(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o) + \mathbf{g}_y(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o). \quad (*)$$

Derive (*) by regarding the graph of $\mathbf{g}(\mathbf{x}, \mathbf{y})$ as a level surface.

Q10.4. Consider the function $f(x, y) = e^{2x+y}$.

(a) Find the linearization of f at the point $(0, 0)$.

(b) Find the linearization of f at the point $(1, 2)$.

[Remark: In both (a) and (b), give your answers in the form $ax + by + c$.]

Q10.5. Consider the function $w = xy^2z^3$. Find the (total) differential dw in terms of dx , dy and dz .

Q10.6. Consider the function $w = w(x, y, z)$ defined implicitly by the equation

$$z^2 + xyw + yw^3 = 0.$$

Find the differential dw in terms of dx , dy and dz .

[Suggestion: Regard both sides as functions of x, y, z, w , and take differentials on both sides. Then solve for dw .]

[Remark. Alternatively, you may also use implicit differentiation to find w_x, w_y, w_z and then get dw .]

Q10.7. Consider the function $f(x, y, z) = xz + \sin(xyz)$.

Using linear approximation, estimate the change in the value of f when (x, y, z) changes from $(\frac{1}{2}, \pi, \frac{1}{3})$ to

$(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02)$.

Q10.8. Let $f(x, y)$ be a function, and (x_o, y_o) be a point in the domain of f . Consider the quadratic polynomial (in two variables) given by

$$\begin{aligned} g(x, y) = & a + b(x - x_o) + c(y - y_o) \\ & + d(x - x_o)^2 + e(x - x_o)(y - y_o) \\ & + f(y - y_o)^2, \end{aligned}$$

where a, b, c, d, e, f are constants.

Determine a, b, c, d, e, f (in terms of data on f) if

$$\begin{aligned} g(x_o, y_o) &= f(x_o, y_o), & g_x(x_o, y_o) &= f_x(x_o, y_o) \\ g_y(x_o, y_o) &= f_y(x_o, y_o), & g_{xx}(x_o, y_o) &= f_{xx}(x_o, y_o), \\ g_{xy}(x_o, y_o) &= f_{xy}(x_o, y_o), & \text{and } g_{yy}(x_o, y_o) &= f_{yy}(x_o, y_o). \end{aligned}$$

Hence write down $g(x, y)$.

[Remark: $\mathbf{g}(\mathbf{x}, \mathbf{y})$ give a quadratic approximation of $\mathbf{f}(\mathbf{x}, \mathbf{y})$ for $(\mathbf{x}, \mathbf{y})$ near $(\mathbf{x}_o, \mathbf{y}_o)$. Also, $\mathbf{g}(\mathbf{x}, \mathbf{y})$ is called the 2^{nd} -degree Taylor polynomial of $\mathbf{f}$ at $(\mathbf{x}_o, \mathbf{y}_o)$.]

Q10.9. Consider the function $\mathbf{f}(\mathbf{x}, \mathbf{y}) = \mathbf{e}^{\mathbf{x}+4\mathbf{y}} \cos \mathbf{x}$.

(a) Find the second degree Taylor polynomial of $\mathbf{f}$ at $(\mathbf{0}, \mathbf{0})$.

(b) Using part (a), find an approximate value of $\mathbf{f}(\mathbf{0.2}, \mathbf{0.1})$.

Q11.1. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, y) = \mathbf{x}y$ subject to the constraint

$$(\mathbf{x} - 2)^2 + y^2 = 4.$$

Q11.2. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, y) = \mathbf{x}y$ subject to the constraints

$$(\mathbf{x} - 2)^2 + y^2 = 4 \quad \text{and} \quad \mathbf{x} \leq 2. \quad (*)$$

[Remark. The above question is modified from Q11.1 by adding an additional constraint $\mathbf{x} \leq 2$. So you may find your calculations in Q11.1 useful.]

Q11.3. Find the point(s) on the sphere $x^2 + y^2 + z^2 = 1$ furthest from the point $(2, 1, 2)$.

Q11.4. (Q08.5. revisited) A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from materials costing 10 dollars per square meter. The two lateral faces are made from materials costing 5 dollars per square meter. The bottom face is made from materials costing 30 dollars per square meter. Using the method of Lagrange multipliers, find the minimum material cost of the rectangular box if the volume of the box is 6 cubic meters. Give also the dimensions (i.e. length, width, and height) of such box.